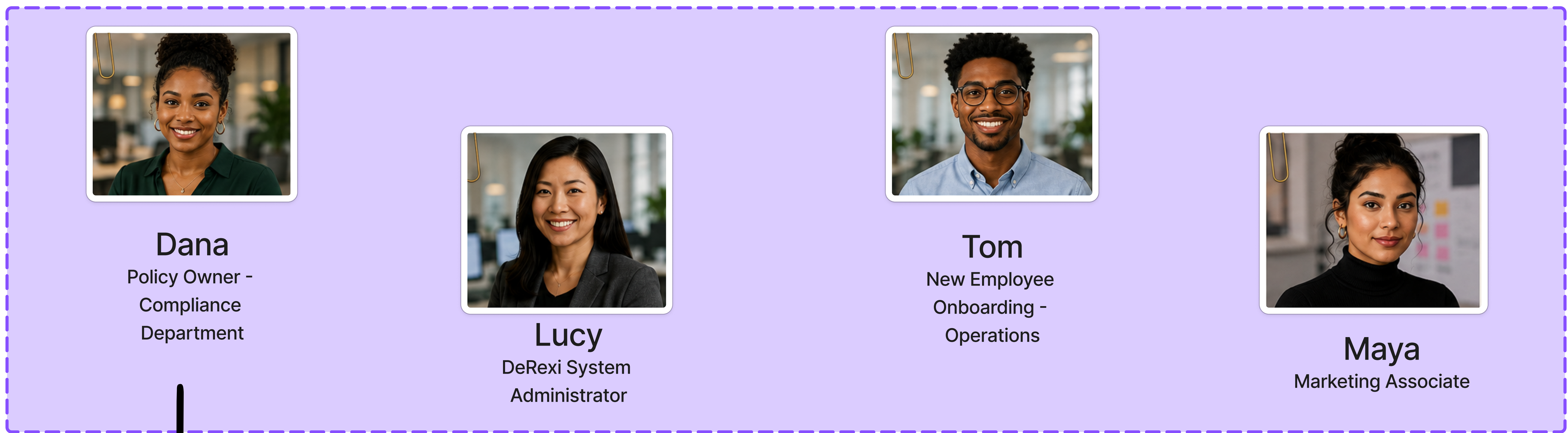


DeRexi Core System Components



It is important that the **TCP method** is used at the Transport layer because Dana is accessing company policies and potentially updating important information. We want the information to arrive completely and correctly, not simply as fast as possible.

<u>OSI Model</u>	<u>What it means for Dana's laptop</u>	<u>TCP/IP Model</u>
7. Application	Dana uses a browser to open DeRexi	Application
6. Presentation	Data is formatted and encrypted, such as with TLS	Application
5. Session	Maintains Dana's active login/session	Application
4. Transport	TCP reliably delivers the data	Transport
3. Network	Dana's laptop uses an IP address and packets are routed	Internet
2. Data Link	Her laptop uses its MAC address to communicate on the local network	Network Access
1. Physical	Wi-Fi or Ethernet physically carries the signal	Network Access

<u>Setting</u>	<u>Example</u>	<u>What Does This Mean</u>
Network	192.168.10.0/24	192.168.10.0 represents the subnet itself.
Subnet Mask	255.255.255.0	Defines which portion of the IP identifies the network and which portion can identify individual devices.
Dana's Laptop IP	192.168.10.25	This is Dana's individual address inside that network.
Default Gateway	192.168.10.1	This is the door out of the subnet. If Dana needs to communicate with something outside her local network, her traffic goes toward this gateway.
Usable device range	192.168.10.1 – 192.168.10.254	IP addresses within this range are assigned to other devices under the Compliance department's network subnet.
Broadcast Address	192.168.10.255	This is a special address used to communicate with all devices within Dana's subnet.

HTTPS + TLS 1.3

HTTPS = Hypertext Transfer Protocol Secure
TLS = Transport Layer Security

Dana accesses DeRexi through: HTTPS over TCP port 443. TLS encrypts the information traveling between Dana's laptop and DeRexi.

HTTPS/TLS = protects the data while it's moving.

WPA3-Enterprise

WPA = Wi-Fi Protected Access
WPA3 = is the newer generation of Wi-Fi security.

DeRexi would specifically use: WPA3-Enterprise for a corporate environment rather than treating this like somebody's home Wi-Fi.

That helps protect Dana's wireless communication between Dana's laptop and the corporate wireless network

802.1X

This one is an IEEE standard (Institute of Electrical and Electronic Engineers standards).

802.1X = network access authentication

Before Dana's device is allowed onto the corporate network, the organization can require it to authenticate. WPA3 is the tool helping to protect the wireless connection, while 802.1X helps answer: "Is this user/device actually authorized to join our corporate network?"

DNS and DNSSEC

DNS = Domain Name System

Dana types something like: "derexi.company.com" DNS figures out which IP address belongs to that name.

DNSSEC = Domain Name System Security Extensions

Its purpose is to help verify that DNS information hasn't been forged or tampered with.

DHCP

DHCP = Dynamic Host Configuration Protocol

This gives Dana her network configuration automatically: her IP address, subnet mask, default gateway, and DNS server.

So instead of someone manually configuring every employee computer, DHCP handles much of that automatically. Dana's IP address is still 192.168.10.25 but the company manages IP addresses through DHCP.

VPN

VPN = Virtual Private Network

VPN will be needed for remote work access. Instead of allowing Dana to directly connect to internal company resources over an distrusted internet connection, she establishes an encrypted connection back to the organization.

There are different VPN technologies, DeRexi could use IPsec and TLS-based VPNs.

IPsec = Internet Protocol Security

